

doc link:-

<https://docs.google.com/document/d/11DZQ8raX94P4AZ2b3f3GpPLzK5fl4-nwV1d9g0TAEvE/edit?usp=sharing>

chapter link:- <https://jax-ml.github.io/scaling-book/applied-training/>

The LLaMA-3 model family

[1]

includes 3 main models: LLaMA 3 8B, 70B, and 405B. We'll mostly focus on 70B, and leave 8B and 405B for you to explore in the problem section at the end. Here's the architecture for LLaMA 3-70B, taken from the LLaMA [HuggingFace page](#).

hyperparam	value
(L)	80
(D)	8,192
(F)	28,672
(N)	64
(K)	8
(H)	128
(V)	128,256

Question: From this table, can we calculate the LLaMA 3-70B parameter count? 🤔
Let's apply the content of [Section 4](#) and see if we can get 70B!

param	formula	count
FFW params	$d_{\text{model}} * d_{\text{ff}} * 3$ (for SwiGLU gate, up, and down projections) $* n_{\text{layers}}$	$8,192 * 8,192 * 3.5 * 80 = 56.3e9$
Vocab params	2 (input and output embeddings) $* n_{\text{embeddings}} * d_{\text{model}}$	$2 * 128,256 * 8,192 = 2.1e9$
Attention params	$n_{\text{layers}} * [2$ (for q embedding and concatenated output projection) $* d_{\text{model}} * n_{\text{heads}} * d_{\text{qkv}} + 2$ (for k and v) $* d_{\text{model}} * n_{\text{kv_heads}} * d_{\text{qkv}}]$	$80 * (2 * 8,192 * 64 * 128 + 2 * 8,192 * 8 * 128) = 12e9$

$$56.3e9 + 2.1e9 + 12e9 = 70.4e9$$

My Answer:-

Vocab to emb and emb = $2 \cdot D \cdot V = 2 \cdot 128256 \cdot 8192 = 2.1B$ total

kqv embed to heads and output of attention(NHD) = $2 \cdot D_{NH} + 2 \cdot D_{KH}$ per layer = $2 \cdot D_H(N+K) = 2 \cdot 8192 \cdot 128 \cdot (64+8) = 0.151 B$ per layer = 12.1B total

MLP per layer = $D_F \cdot 3 = 3 \cdot 8193 \cdot 28672 = 0.7B$ per layer = 56.37B total

Question: How many FLOPs does LLaMA-3 perform per token per training step? *This helps us determine how expensive the whole training process will be.*

My Answer:- rule of thumb for training the num of flops is $6 \cdot \text{num token} \cdot \text{param count}$ which is here around 420B FLOPs

Question: LLaMA 3 was trained for about 15 trillion tokens. How many FLOPs is that total?

My Answer:-

it is around $6 \cdot 1.5e13 \cdot 7e10 = 6.3e24$ Flops

Question: Let's say we wanted to train on a full TPU v5p pod with $16 \times 20 \times 28 = 8960$ chips. How long would this take to train at 40% MFU in bfloat16, assuming we are compute-bound?

My Answer:-

TPU v5p per chip flop/s is $4.59e14$

so the amount of time is around $(1/0.4) \cdot (6.3e24 / (8960 \cdot 4.59e14)) = 3.83e6$ seconds = 44 days

Question: LLaMA 3-70B was pretrained with a batch size of about 4M tokens. How many TPUs do we need at minimum to train with this batch size? *You can assume bfloat16 parameters and float32 optimizer state, and that you checkpoint gradients 4 times per layer.*

My Answer:-

total parameter in bf16 = 70B = 140GB

for optimiser of f32 = $2 \cdot 2 \cdot (140) = 560GB$

memory for check point per layer param after all norm and activation bytes = $2BD \cdot 4$ per layer

so total is $8 \cdot BD \cdot L = 8 \cdot 4e6 \cdot 8192 \cdot 80 = 21 TB$

per chip HBM is 96 GB so minimum we require is $21.7\text{TB}/96\text{GB}=230$ chips at minimum

Question: Under the same assumptions as the question above, if we use 8960 TPU v5p chips, how much memory will we use per-chip?

My answer —

per chip memory= 2.42GB

Let's stick to our setting from above and say we want to train LLaMA 3-70B with 4M token batch size (1024 sequences of length 4096 per batch) on a TPU v5p pod of 8960 chips. Let's discuss what the best sharding strategy is for this model.

Question: Under the assumptions above, can we train our model with FSDP alone? To start, let's say we can't do any sequence/context parallelism. *This should be the first idea you have, since it's simple and will introduce no extra communication if it works.*

My answer:-

if we don't do sequence parallelism then we have 1024 sequences and one tpu chip do processing for 1 sequence so we won't be able to use nearly 8000 chips so it's definitely waste

Question: Let's relax the requirement of not doing any sequence sharding. If we allow ourselves to do FSDP over both the batch *and* sequence axes, can we train LLaMA 3-70B with only FSDP on 8960 chips?

My answer:-

we can surely fit the model params in our HBM as calculated in last question

now for compute bound we need $B/N > c/w_{\text{chip}}/3 \Rightarrow B > 8960 * 2550 / 3 = 7.6\text{M}$

so we cannot use pure FSDP for this batch size but combination of FSDP and magatron may work as for it we require

$B/N > \alpha^2 / 2F$

$B > N * 2550 * 2550 / (2 * 28672)$

so $B > 8960 * 2550 * 2550 / 2 / 28672 = 1\text{M}$

so we can use mixture of them

Question: Now let's look at mixed tensor parallelism and FSDP. Does there exist some combination that lets us remain compute-bound? What amount of FSDP and tensor parallelism should we do if so?

My answer:-

partial answer in the above answer for that we need the value of X to be $\sqrt{BM \times N / FMy}$

so $X_{opt} = \sqrt{4.19e6 \times 2 \times 8960 / 28672} = 1619$

or we can do 809 X_{opt} if we switch $Mx=1$ and $My=2$

so nearest multiple of 2 is 512 = X_{opt} or 2048 both works depending upon mesh axis we select

Question 1 [Scaling LLaMA 70B to more chips]: say we want to train LLaMA 3-70B on 4 pods with the same batch size. What parallelism scheme would we use? Would we be compute or communication bound? Roughly how long would it take to train? Make sure to use the correct roofline bound.

My Answer:-

we will do pure data parallelism across the pods which is 1M tokens per pod then internally we will do mixed fsdb and megatron

in multi pod we need $B/slices > C/Wdcn$ to be compute bound

so batch per slice > 71000 so $1M > 71000$ so we are compute bound

now to check for each pod for compute bound in mixed scenario

as calculated we need $B > 8960 \times 2550 \times 2550 / 2 / 28672 = 1M$ so we are barely compute bound

for amount of time required it will be $6 \times \text{num tokens} \times \text{model size} / (4 \times \text{num chips} \times \text{per chip flops})$

so time = $6 \times 15e12 \times 7e10 / (4 \times 8960 \times 4.69e14) = 3.75e5$ seconds = 4-5 days at full efficiency

at 40% efficiency its 11 days

Question 2 [LLaMA 405B]:

(a) Using the LLaMA 3-405B [config](#), write a table with all the key hyperparameters as above. How many total parameters does this model have? How many FLOPs per training step? How many FLOPs do we perform if we train for 15T tokens?

(b) Assume we want to train on 8 TPU v5p pods. What parallelism scheme would we use? How long would training take? Would we be compute or comms bound?

My answer:-

N layers (L)	126
D model (D)	16384
dff (F)	53248
n heads(N)	128
n kv heads (K)	8
dqkv (H)	128?=D/N
n embedding (V)	128256

number of parameters:-

vocab = $2 \cdot d \cdot v = 4.2B$

kqvo = $(2 \cdot DNH + 2 \cdot DKH) = 2 \cdot DH(N + K) = 0.55B$ per layer = 69.76B total

MLP layer = 3DF per layer = 2.617B per layer = 329.77B total

total = 404B added

for per token step num of flops = $6 \cdot$ total param flops = 2.43 TFLOPs

to train on 15T tokens its 36.45 e24 FLOPs operations required

(b) 1st we will use pura data parrallelism for on 8 pods

for that as calculated for per pod we required $B > 7.6M$ per pod

to escape DCN comune bound

in mixed scenario we wqill require $B/N > \alpha^2/2F$

so in this case $B > 8960 \cdot 2550 \cdot 2550 / (2 \cdot 53248)$ $B > 0.5M \Rightarrow$ total batch size spreaded per pod $> 0.5M$ and total $> 4M$

we cannot go for normal FSDP which require around $B/N > c/wici \Rightarrow B > 22M$

for magatron case we can go for $F/Y > c/wici/3$ lets check this $F > 7.6M$ which is not the case

so we can go for mixed sharding and full data parrallelism in the pods